

DINESH SEKAR

Solution Architect | Infrastructure and Network Automation

Enterprise DDI • Hybrid Cloud • IT Infrastructure • Agentic AI

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Professional Summary

Solution Architect and infrastructure technical leader with 16+ years of experience designing, modernizing and managing enterprise IT infrastructure. Specializes in Infoblox DNS, DHCP and IPAM, hybrid AWS/Azure architecture, network automation and AI-enabled operations. Translates business and service requirements into solution designs, HLD/LLD, security assessments, implementation plans and operational standards.

Led modernization of 700,000+ DNS records and architected global DHCP migration covering 10,000+ scopes across 10+ international sites. Completed end-to-end architecture for a hybrid IPAM platform using a web dashboard, PostgreSQL and Agentic AI/MCP; built a custom dig-based DNS diagnostics and troubleshooting tool.

Experienced in end-to-end IT infrastructure ownership across networking, servers, virtualization, identity, cloud and service delivery. Responsibilities include IT budget planning, vendor management, licensing and cost optimization, alongside technical leadership and mentoring of 10+ engineers.

Architecture and Engineering Highlights

- Global DHCP: **10,000+** migrated scopes, 10+ international sites and centralized Infoblox Grid management in Sydney.
- Enterprise DNS: **700,000+** records migrated to Infoblox with zero downtime.
- Hybrid IPAM: End-to-end platform architecture spanning UI, backend, PostgreSQL, heterogeneous sources and MCP.
- DNS diagnostics: Custom dig tool and an agentic AI analyser with reusable skills for layer-by-layer troubleshooting.
- Engineering impact: Mentored 10+ engineers; Infoblox provisioning automation reduced manual workload by **70%**.

Core Competencies

DDI and cloud DNS: Infoblox NIOS, Grid management, HA, DNS, DHCP, IPAM, Windows DNS/DHCP, AWS Route 53, Azure DNS and domain governance

Architecture and leadership: Solution design, HLD/LLD, security assessments, migration planning, platform lifecycle, stakeholder engagement and engineering mentoring

Automation and platforms: Python, Ansible, Terraform, CloudFormation, FastAPI, ReactJS, PostgreSQL, APIs, AWS Lambda and Step Functions

AI and service delivery: Agentic AI, MCP, Claude, skill-guided diagnostics, IPAM forecasting, Jira, ServiceNow and Confluence

Infrastructure management: Windows Server, Active Directory, VMware ESXi, FortiGate, VPN, Office 365, IT budgeting and vendor management

Professional Experience

Commonwealth Bank of Australia

Senior System Engineer

January 2022 to Present

Functional focus: Enterprise DDI architecture, DNS, DHCP and IPAM governance and AI network automation

Global DDI Architecture and Modernization

- Architected a global Infoblox DHCP solution with centralized Grid management in Sydney and distributed Grid Members across **10+ international** sites.
- Led architecture and migration of **10,000+** DHCP scopes from Windows DHCP to Infoblox, aligning the target platform with geographically distributed service requirements.
- Led enterprise DNS modernization from Windows/Linux DNS to Infoblox, migrating **700,000+** resource records with zero downtime and preserving business continuity.
- Designed and implemented hybrid Infoblox architecture across on-premises, AWS and Azure, improving DNS query performance and separating workload environments.
- Owned Infoblox platform lifecycle across NIOS upgrades, licensing, HA/Grid management, backups and critical patching; aligned operational standards with security and architecture teams.

Hybrid IPAM Platform and Agentic AI

- Completed end-to-end solution architecture for centralized hybrid IPAM visibility across Infoblox, Windows, cloud IPAM, Linux and other infrastructure data sources.
- Designed the frontend dashboard, backend services, PostgreSQL data layer and integration architecture to consolidate IPAM information into a single pane of glass.
- Designed an agentic AI/MCP layer for natural-language retrieval of IPAM data from authoritative systems, including integration paths to ServiceNow, Jira and other operational tools.
- Architected and delivered a Python/FastAPI and ReactJS DDI operations platform incorporating DNS diagnostics, SSL validation, IPAM utilization and forecasting, and DHCP monitoring.
- Implemented Claude/MCP incident diagnostics across Infoblox, Windows DNS, AWS Route 53 and Azure DNS APIs for resolution tracing, record validation, reverse lookups and service reachability checks.

Custom DNS Diagnostics and Troubleshooting

- Designed and built a custom dig-based DNS tool supporting DNS record type validation, bulk DNS lookup and reverse DNS lookup, bringing common diagnostic workflows into one interface.
- Developed an agentic AI DNS troubleshooting analyser using reusable skills to guide deep, orchestrate layer-by-layer analysis and interpret diagnostic results for incident investigation.

Infoblox SSL Certificate Lifecycle Automation

- Designed, architected and implemented end-to-end serverless SSL certificate automation for Infoblox Grid Masters using AWS Lambda, S3 and venafi certificate replacing manual certificate installation with automated renewal and deployment.

Automation Governance and Service Ownership

- Implemented infrastructure provisioning with Terraform, Ansible and AWS CloudFormation; built Infoblox playbooks for DNS record, subnet and zone provisioning, reducing manual workload by **70%**.
- Architected and delivered serverless SSL certificate provisioning and renewal with AWS Step Functions and Lambda, reducing repetitive manual operations and dedicated infrastructure requirements.

- Governed internal DNS on Infoblox/Windows and external DNS on Route 53/Azure DNS, including CSC registrar domain lifecycle, billing, compliance and unused-domain decommissioning.
- Oversaw incident escalation and service delivery through Jira and ServiceNow, supporting more than 99.9% DNS availability and standardizing operational processes.

Architecture Governance and Technical Leadership

- Produced HLD/LLD, security risk assessments, implementation plans, deployment documentation and runbooks; maintained architecture knowledge in Confluence and approved designs on GitHub static pages.
- Led and mentored 10+ engineers across DNS, DHCP, IPAM and cloud domains, providing technical guidance, prioritization and knowledge transfer; collaborated on governance, RBAC and audit compliance.

Excelacom Technologies

Senior Associate IT

July 2019 to January 2022

Enterprise Infrastructure and Hybrid Cloud

- Managed end-to-end enterprise IT infrastructure spanning network, cloud and on-premises platforms, with responsibility for service reliability, performance and operational coordination.
- Supported and implemented AWS/Azure hybrid environments, connecting cloud capabilities with enterprise infrastructure requirements.
- Administered VMware ESXi and Windows Server environments, maintaining platform availability, resource utilization and system performance.
- Managed Active Directory, DNS, DHCP and Group Policy services to support enterprise identity, connectivity and endpoint configuration.
- Managed FortiGate firewalls, IPsec/SSL VPN and high availability configurations to support secure network access.
- Led the migration of on-premises Exchange to Office 365, coordinating the transition of enterprise collaboration services.
- Coordinated onshore/offshore teams and external vendors for infrastructure delivery, issue resolution and ongoing service support.

TPF Software TSI Chennai

Senior Lead IT Infrastructure

September 2013 to July 2019

Enterprise Infrastructure and Hybrid Cloud

- Led end-to-end IT infrastructure operations across three regions, covering network, cloud and on-premises platforms; coordinated onshore/offshore teams, service delivery and incident management.
- Designed and implemented cloud and DevOps strategies across AWS and Google Cloud, including workload migration and hybrid connectivity.
- Managed FortiGate firewalls, VPN connectivity and security controls including IPS, web filtering and application control.
- Owned IT budget planning, vendor management and cost optimization initiatives, aligning infrastructure spend and service requirements.
- Oversaw enterprise identity and network services including Active Directory, DNS, DHCP and GPOs, alongside VMware ESXi and Windows Server administration.

Dot Com Infoway

System Administrator

February 2013 to August 2013

- At Dot Com Infoway, administered AD DS, DNS, DHCP, Windows Server, clustering, IIS and hosting services

ZTCS

System Administrator

June 2010 to February 2013

- . At ZTCS, provided LAN, client systems, Outlook, hardware and application support.

Certifications and Credentials

- Infoblox Core DDI Configuration and Administration (CDCA)
- AWS Certified Solutions Architect – Associate
- Microsoft Azure Solution Architect; Microsoft Azure Administrator
- Microsoft Certified Solutions Associate (MCSA) Windows Server 2012
- Microsoft Certified IT Professional (MCITP) Enterprise Administrator
- Infoblox Champion Silver

Education

- M.Sc. Cyber Forensics and Information Security | Madras University | 2018–2020
- B.Sc. Information Technology | Alagappa University, Distance Education | 2015–2017\
- DCE | Little Flower Polytechnic College | 2007–2010